Week-8

Aim: Simulate a cloud scenario using CloudSim and run a scheduling algorithm in CloudSim.

Simulation:

- Simulation is making a virtual model of a real system to see how it would act in different situations.
- It helps us test and understand systems safely without changing the real thing.

CloudSim:

- CloudSim is an open-source framework, which is used to simulate cloud computing infrastructure and services.
- It is developed by the CLOUDS Lab organization and is written entirely in Java.
- It is used for modelling and simulating a cloud computing environment

Some of the most common classes used during simulation are:

- **Datacenter:** used for modelling the foundational hardware equipment of any cloud environment, that is the Datacenter.

- **Host:** this class executes actions related to management of virtual machines. It also defines policies for provisioning memory and bandwidth to the virtual machines, as well as allocating CPU cores to the virtual machines.

- **VM:** this class represents a virtual machine by providing data members defining a VM's bandwidth, RAM, mips (million instructions per second), size while also providing setter and getter methods for these parameters.

- **Cloudlet:** a cloudlet class represents any task that is run on a VM, like a processing task, or a memory access task, or a file updating task etc.

- **CloudSim:** this is the class responsible for initializing and starting the simulation environment after all the necessary cloud entities have been defined and later stopping after all the entities have been destroyed.

Benefits of CloudSim:

- Open source and free of cost
- Easy to use and Scalable.
- Risks can be evaluated at an earlier stage.
- Provides pre-defined simulation packages

Scheduling Algorithm:

In CloudSim, a scheduling algorithm schedule how cloud resources, like virtual machines (VMs), are assigned to different tasks or jobs.

Steps:

1. Java Installation

a. Check java in your system

Open Command Prompt (Windows) or Terminal (Mac/Linux).

Type the following command to check the Java version:

java -version

If Java is installed, you'll see the version details.

If not installed, proceed to the next step to download and install Java.

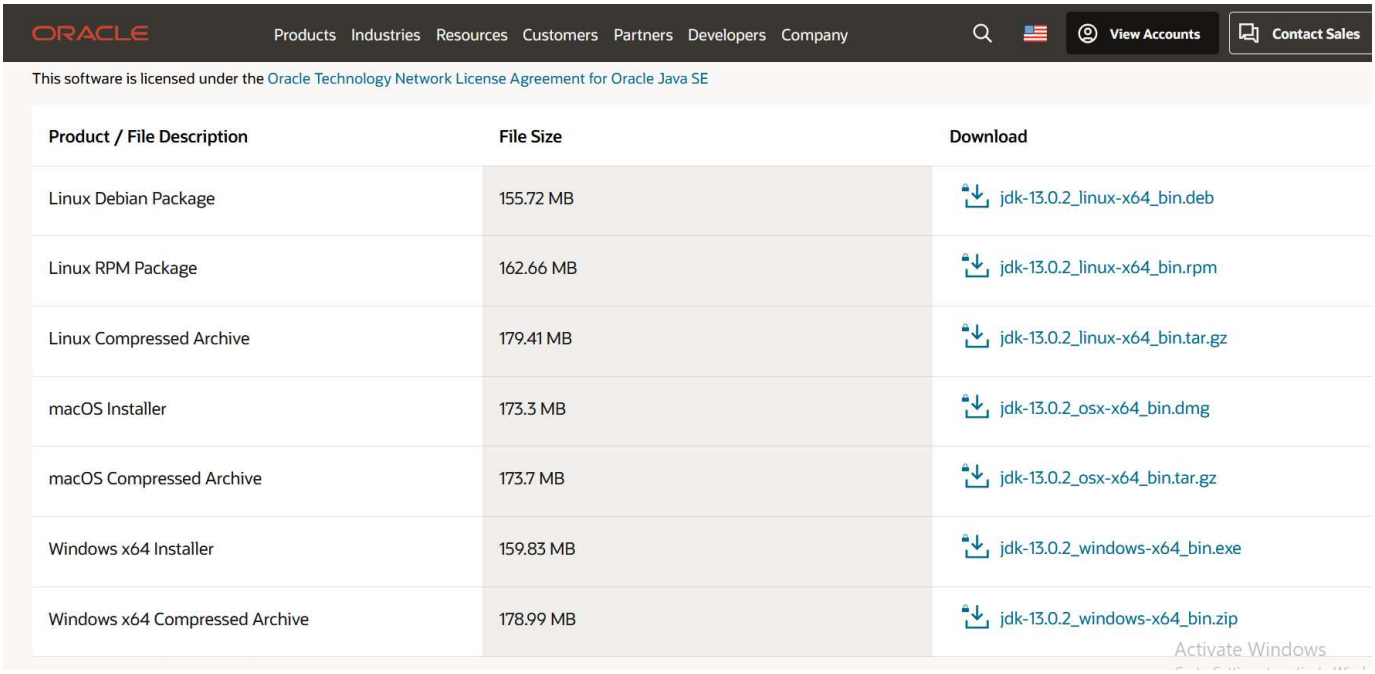
Visit the official Java download page.

b. Download the appropriate version for your operating system (64-bit)

For Windows, download the **.exe** file.

For Mac, download the **.dmg** file.

For Linux, download the **.tar.gz** file.

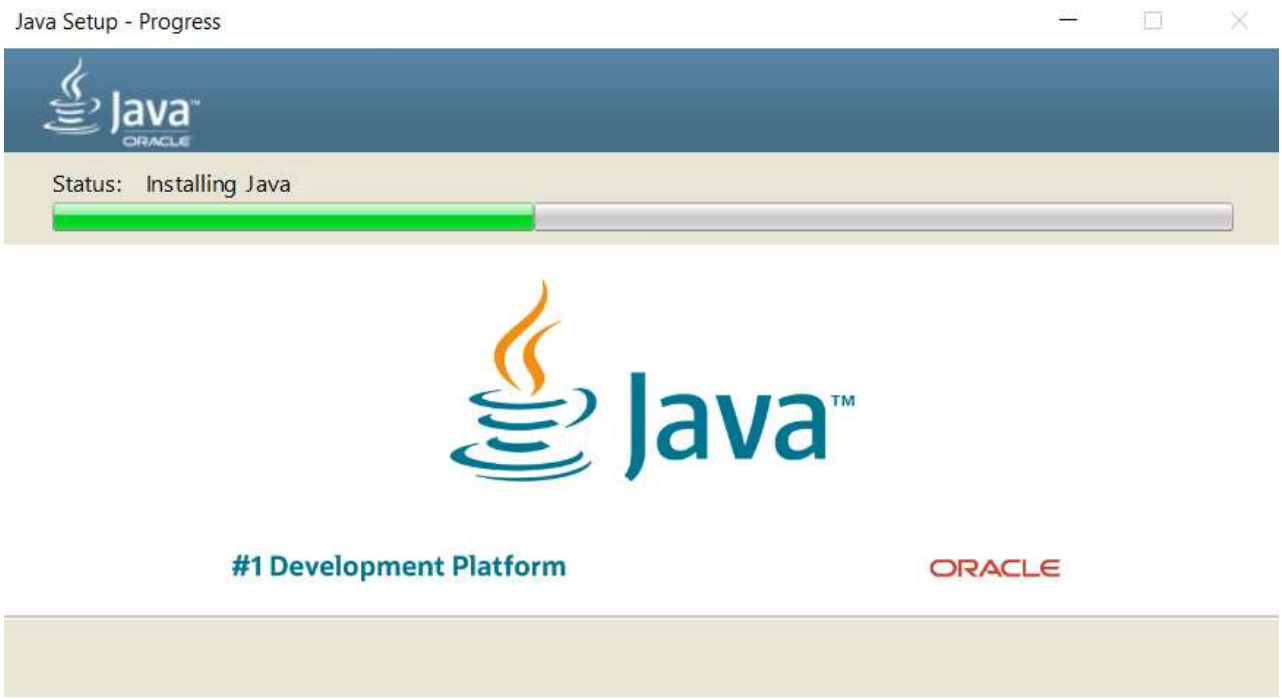


https://www.oracle.com/java/technologies/javase/jdk13-archive-downloads.html

c. Install Java

Run the downloaded setup and follow the installation instructions.

Complete the installation process.



d. Set Path Variable for Java (Windows)

Right-click on This PC or My Computer and go to Properties.

Select advanced system settings.

Click Environment Variables.

In System variables, find Path, select it, and click Edit.

Add the path of the bin folder in the Java installation directory.

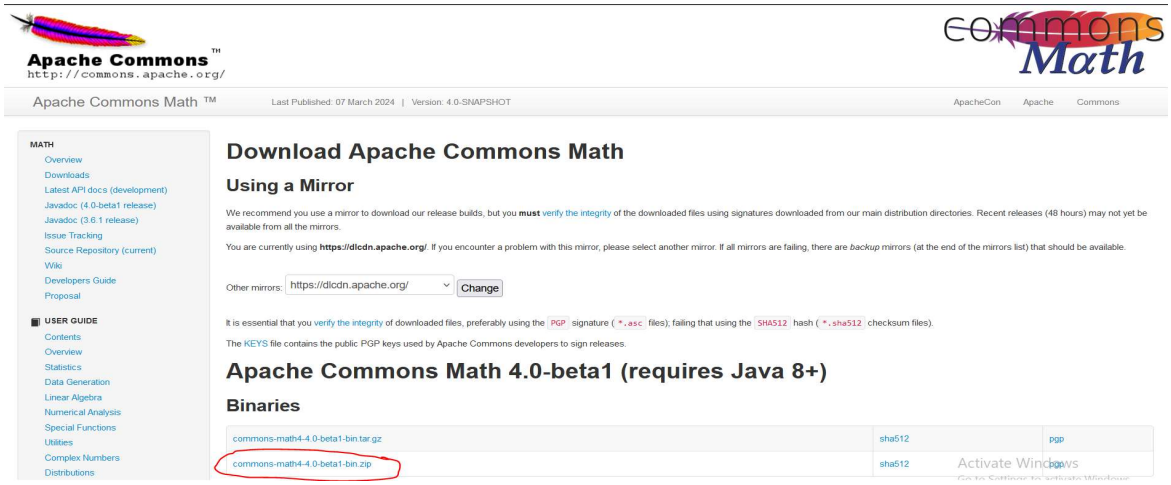
C:\Program Files\Java\jdk-14\bin //Click OK to save the changes.

2. Download CloudSim and additional jar file

a. Download CloudSim

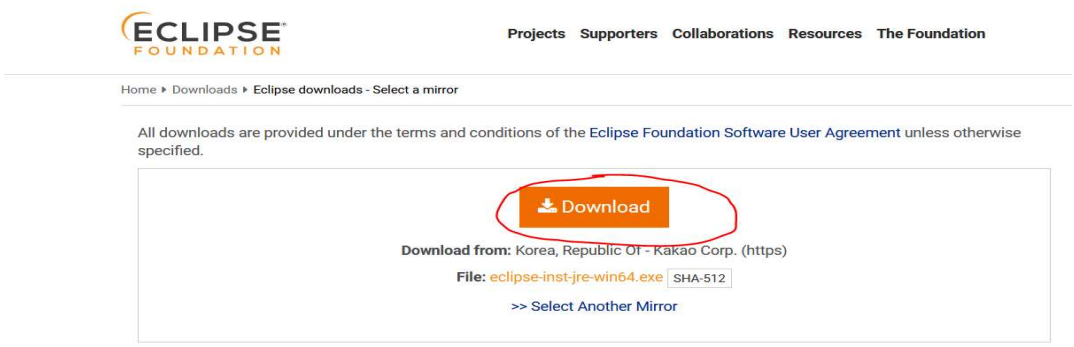
Download CloudSim and extract the contents.

b. Download Commons math 3 jar file

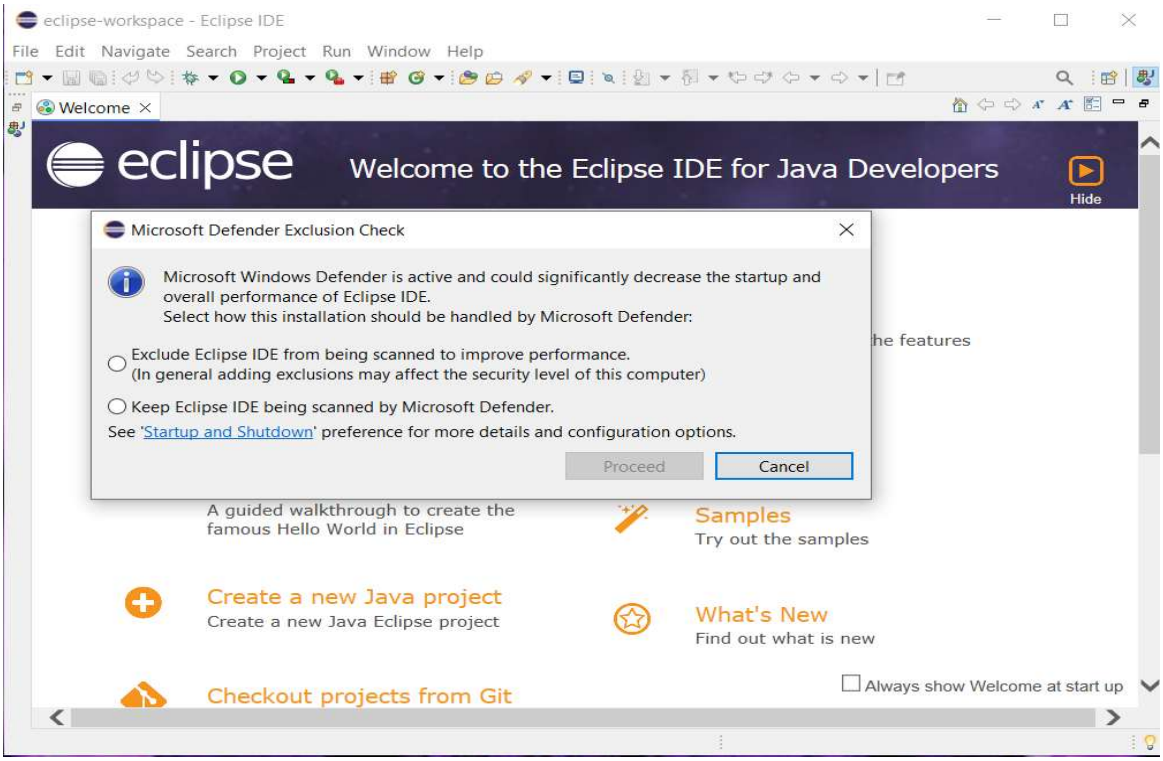


3. Eclipse IDE Installation

- If java 64 bit Installed then download 64 bit Eclipse.

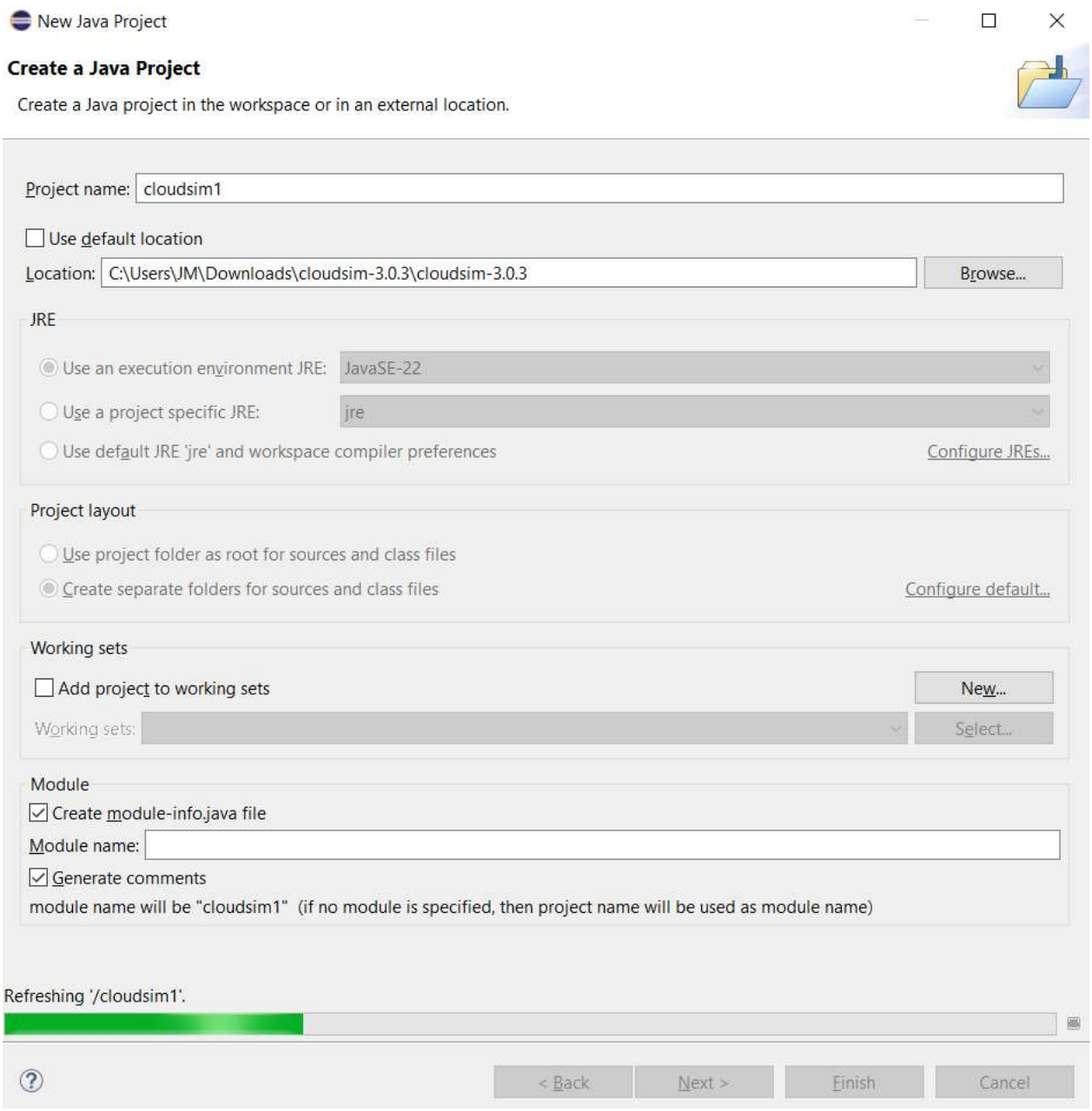


- Install Eclipse IDE

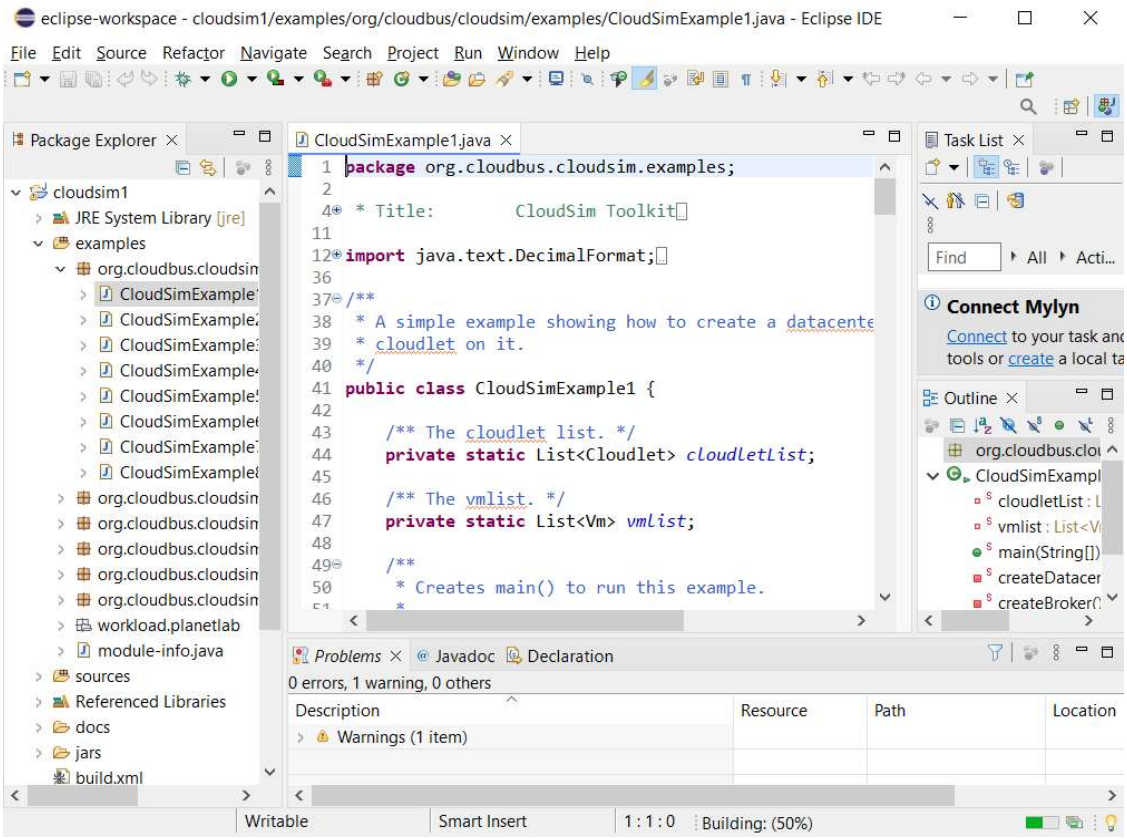


4. Run CloudSim in Eclipse

- Put up commons-math jar file into jar folder of Cloudsim3.0.3
- Build a new java project with Cloudsim3.0.3 folder
  Open Eclipse and create new java project



After creation of our project execute any example from the cloudsim examples



example CloudSim program: Creating a Datacenter, VM and Simulation

```
package org.cloudbus.cloudsim.examples;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.Vm;
public class SimpleVmCreation {
    public static void main(String[] args) {
        try {
            // Initialize CloudSim
            CloudSim.init(1, null, false);
            // Create a VM
            Vm vm = createVm();
            // Print VM information
            System.out.println("VM created: " + vm.getId());
```

```
            // Start the simulation (although no datacenter is created)
            CloudSim.startSimulation();
            CloudSim.stopSimulation();
            System.out.println("Simulation finished!");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
    private static Vm createVm() {
        // Create and return a VM with specified parameters
        return new Vm(0, 0, 1000, 1, 512, 1000, 10000, "Xen", null);
    }
}
```

VM Parameters:

ID: Unique identifier for the VM.

Number of Cores: Set to 1.

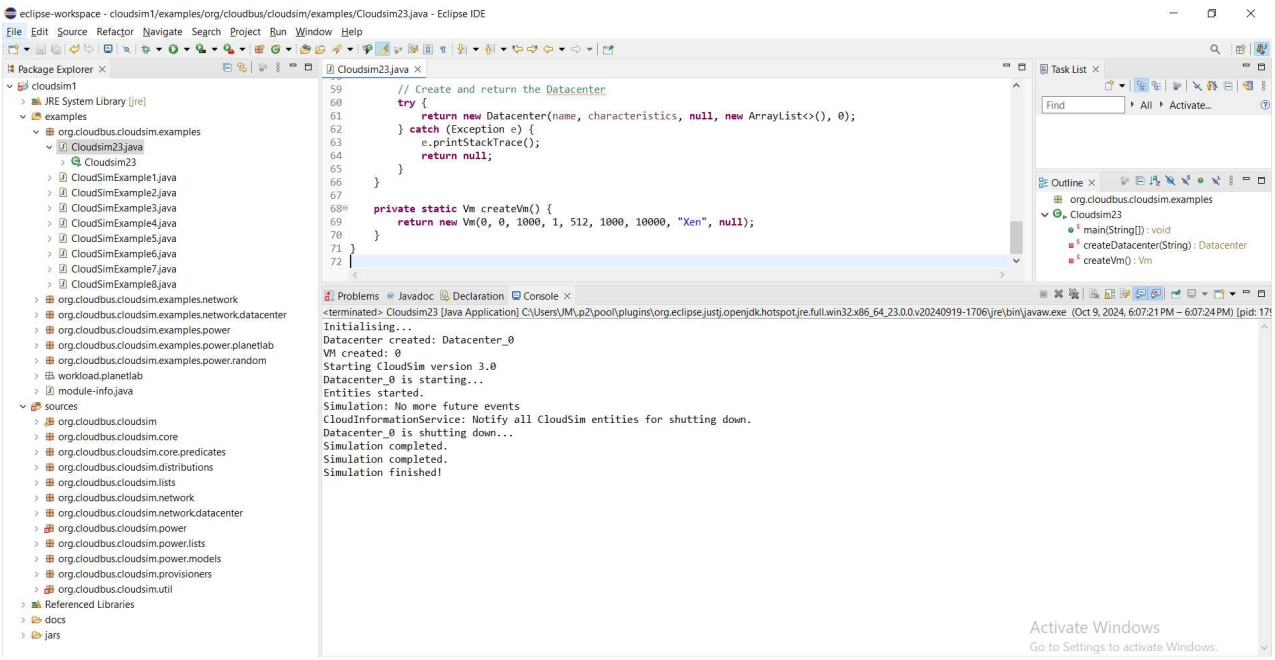
MIPS (Million Instructions Per Second): 1000 (processing power).

RAM: 512 MB.

Bandwidth: 1000 Mbps.

Storage: 10,000 MB.

Hypervisor: "Xen" (type of virtualization).



Output:

```
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Edge0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0 VM is created on 0
1 VM is created on 1
2 VM is created on 2
3 VM is created on 3
4 VM is created on 0
5 VM is created on 1
6 VM is created on 2
7 VM is created on 3
app0
app1
app2
app3
app4
app5
app6
app7
app8
app9
app10
app11
app12
```

```
app13
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Edge0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.

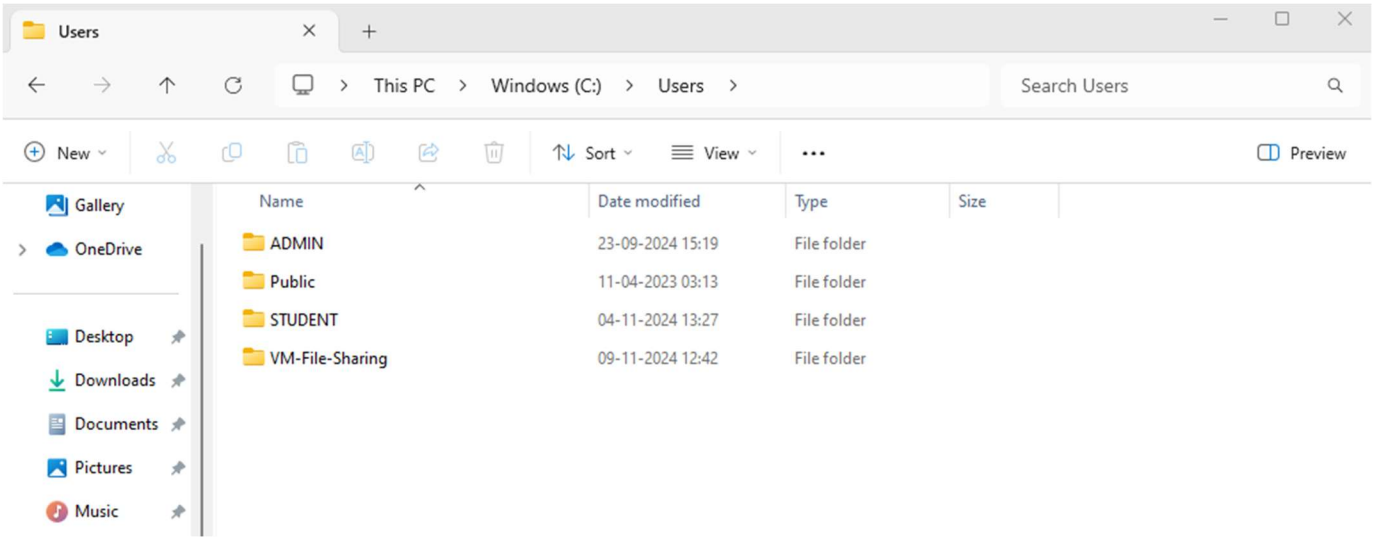
========== OUTPUT ==========
Cloudlet ID STATUS Data center ID VM ID Time Start Time Finish Time
1 SUCCESS 2 0 800 0 800
4 SUCCESS 2 0 800 0 800
7 SUCCESS 2 0 800 0 800
10 SUCCESS 2 0 800 0 800
0 SUCCESS 2 5 800 0 800
3 SUCCESS 2 5 800 0 800
6 SUCCESS 2 5 800 0 800
9 SUCCESS 2 5 800 0 800
13 SUCCESS 2 0 801 800 1601
numberofcloudlet 13 Cached 0 Data transfered 200000
CloudSimExample1 finished!
```

Week-9

Aim: Find a procedure to transfer the files from one virtual machine to another virtual machine.

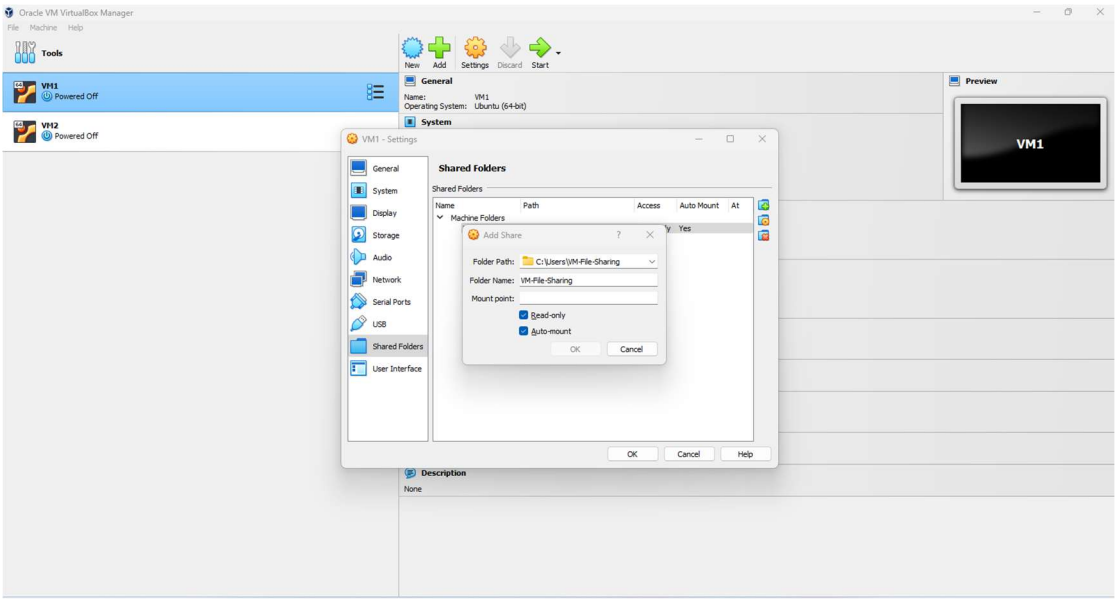
To set up file sharing between your Ubuntu VMs (VM1 and VM2) and the shared folder on your Windows host, follow these steps:

1. First create a Shared folder in C drive



2. Prepare the Shared Folder in VirtualBox Settings:

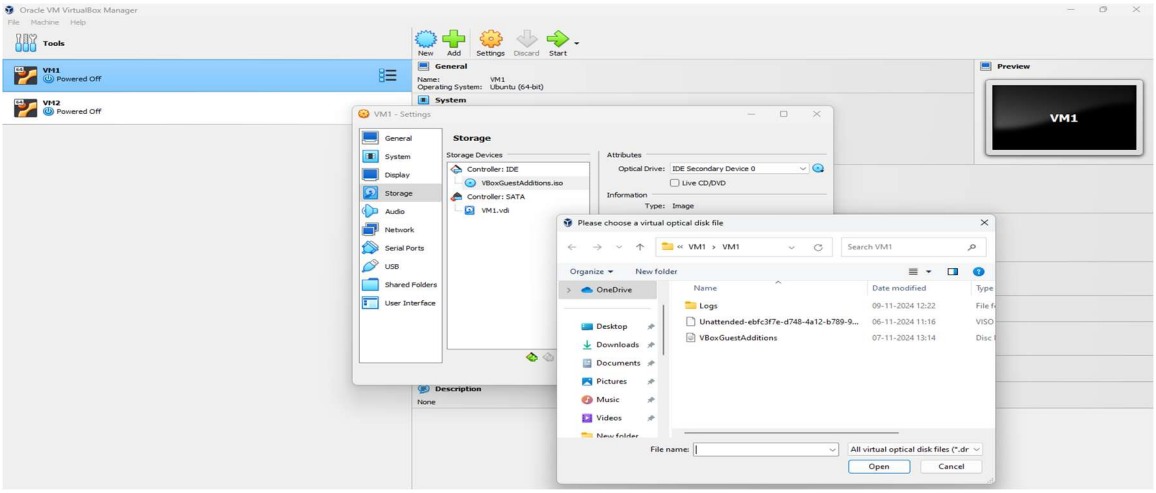
- Shut down VM1 and VM2 if they are running.
- In VirtualBox, select VM1, then go to **Settings > Shared Folders.**
- Click the + icon to add a shared folder.
- For **Folder Path**, select **C:\Users\VM-File-Sharing** on your Windows host.
- Set **Folder Name** (e.g., VM-File-Sharing).
- Check both **Auto-mount** and **Make Permanent** to ensure the folder is accessible each time.



- Repeat these steps for VM2, adding the same shared folder.

3. Install Guest Additions in Both VMs:

- Start each VM (VM1 and VM2) one at a time.
- Open VirtualBox, select VM1 / VM2, and go to Settings.
- Under Storage, make sure that the VBoxGuestAdditions.iso is attached as the optical disk.
- In the Storage settings, find Controller: IDE (or similar).
- You should see VBoxGuestAdditions.iso listed under the optical drive. If it says Empty instead, click it, then select Choose a disk file… and locate VBoxGuestAdditions.iso in C:\Users\STUDENT\VirtualBox VMs\VM2\VM2 (or wherever you stored it).



- Once the ISO is selected, click OK to save the changes.Open a terminal in the VM, then enter these commands to install Guest Additions:

sudo apt update

sudo apt install build-essential dkms linux-headers-$(uname -r)

(This command installs necessary packages for compiling and building software (especially kernel modules like the Guest Additions for VirtualBox).

sudo /media/$USER/VBox_GAs*/VBoxLinuxAdditions.run

(This command installs the VirtualBox Guest Additions, which are required to improve performance and enable features like file sharing and clipboard integration between the host and the VM.)

- Reboot each VM after installation.

4. Access the Shared Folder in Each VM:

- After rebooting, open a terminal in each VM.
- Add the VM user to the `vboxsf` group to access the shared folder:
  sudo usermod -aG vboxsf $USER
- Log out and log back in for the change to take effect.

5. Locate and Use the Shared Folder:

- To access it, navigate with:
  cd /media/sf_VM-File_sharing
- You can now transfer files by placing them in `C:\Users\VM-File_sharing` on Windows, which will automatically be accessible in the `/media/sf_VM-File_sharing` directory on each VM.

